Qiong Zhang *Updated*: October 2026

Contact Info
Division of Computing and Mathematical Sciences
Mohamed bin Zayed University of Artificial Intelligence
Email: qiong.zhang@mbzuai.ac.ae
Homepage: sarahqiong.github.io

Employment

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) 10/2026–present
Abu Dhabi, UAE
Tenure-Track Assistant Professor
Division of Computing and Mathematical Sciences

Renmin University of China 09/2022–09/2026
Beijing, China
Tenure-Track Assistant Professor
Institute of Statistics and Big Data

Education

University of British Columbia 09/2017–05/2022
Vancouver, British Columbia, Canada
Ph.D. in Statistics
Supervisor: Professor Jiahua Chen
Thesis: Inference under finite mixture models: distributed learning and approximate inference

University of British Columbia 09/2015–09/2017
Vancouver, British Columbia, Canada
M.Sc. in Statistics
Supervisor: Professor Jiahua Chen
Thesis: Small area quantile estimation under unit-level models

University of Science and Technology of China 09/2011–06/2015
Hefei, Anhui, China
B.Sc. in Statistics, **School of the Gifted Young**

Research Interests

My research connects statistics and machine learning by developing principled methods for heterogeneous data and modern AI, with an emphasis on scalability, robustness, and trustworthy learning. My main current work focuses on three directions:

- **Mixture Reduction**
  - **Foundations.** Developing optimal-transport methods to approximate mixtures with large numbers of components [18].
  - **Impact Beyond Classical Statistics.** Formulating distributed learning for finite mixture models as a mixture reduction problem to resolve label switching [19] and ensure Byzantine robustness [14], including under partial corruption [5]. Applying mixture reduction to 3D Gaussian Splatting in computer graphics to achieve compaction rates of up to 90% [16].
- **Empirical Likelihood in Modern AI**
  - **A New Federated Learning Paradigm.** Proposing a new federated learning paradigm using empirical-likelihood-based density-ratio modeling to turn the server into an intelligent router and client heterogeneity into an asset [15]. Further generalizing this routing paradigm to dual heterogeneity through semiparametric mixtures [6], and to decentralized federated learning, where clients learn to refer queries to one another at inference time [3].

- **Learning under Label Noise.** Establishing identifiability for Neyman–Pearson multiclass classification without requiring prior knowledge of the noise mechanism [9].

- **Tabular Foundation Models and In-Context Learning**
  - **General-Purpose Statistical Inference.** Interpreting prior-fitted networks as performing amortized Bayesian inference and demonstrating their strong performance across statistical tasks [11].
  - **Beyond Supervised Prediction.** Extending prior-fitted networks to jointly infer the number of clusters and cluster assignments [12] and to perform Thompson sampling for contextual bandits [8].
  - **Limitations and Model Design.** Benchmarking tabular foundation models beyond predictive accuracy [1]. Identifying when repeated measurements hurt tabular foundation models [2] and aligning model architecture with sparse priors [4].

Publications

† denotes corresponding author
* denotes equal contribution or alphabetical order
♢ denotes student author that I supervised/co-supervised for the project

Manuscripts Under Review

1. Tianqi Zhao♢, Jia Guo, Yan Shuo Tan†, and Qiong Zhang†. "TabReliBench: Evaluating tabular foundation models beyond predictive accuracy". Available on arXiv.
2. Yixiang Liu♢, Ruizhe Deng♢, Bibhas Chakraborty, Qiong Zhang, and Yan Shuo Tan. "When repeated measurements hurt tabular foundation models". Available on arXiv.
3. Yuheng Ma*, Chendi Wang*, and Qiong Zhang*. "Learning to refer: Inference-time collaboration in decentralized federated learning". Available on arXiv.
4. Tianqi Zhao*♢, Tianyi Zhuang*♢, Shuo Duan*♢, Guanyang Wang, Yan Shuo Tan†, and Qiong Zhang†. "Architecture alignment with sparse priors in tabular foundation models". Available on arXiv.
5. Yimei Zhang♢, Jiahua Chen, Xiaozhou Wang†, Yan Shuo Tan, and Qiong Zhang†. "Byzantine-tolerant distributed learning of finite mixture models under partial corruptions". Available on arXiv.
6. Zijian Wang♢, Pengfei Li, Guangyu Yang†, and Qiong Zhang†. "FedSPM: Routing-enabled federated learning under dual heterogeneity via semiparametric mixture". Available on arXiv.
7. Ruinan Jin, Beidi Zhao, Myeongkyun Kang, Qiong Zhang, and Xiaoxiao Li. "Verification mirage: Mapping the reliability boundary of self-verification in medical VQA".
8. Yan Shuo Tan, Kenyon Ng, Ruizhe Deng, Sumetha Loganathan, Qiong Zhang, and Bibhas Chakraborty, "PFN-TS: Thompson sampling for contextual bandits via prior-data fitted networks". Available on arXiv.
9. Qiong Zhang, Qinglong Tian, and Pengfei Li. "Neyman-Pearson multiclass classification under label noise". Available on arXiv. (***R&R at JMLR***).
10. Junjun Lang, Qiong Zhang†, and Yukun Liu†. "Minimum Wasserstein distance estimator under covariate shift: Closed form and super-efficiency". Available on arXiv. (***Major revision at JASA***).
11. Qiong Zhang*, Yan Shuo Tan*, Qinglong Tian*, and Pengfei Li. "TabPFN: One model to rule them all?". Available on arXiv. (***Minor revision at JASA***).

Published

12. Tianqi Zhao♢, Guanyang Wang, Yan Shuo Tan, and Qiong Zhang†. "TabClustPFN: A prior-data fitted network for tabular data clustering". **NeurIPS**, 2026.

13. Ruinan Jin[◇], Minghui Chen[◇], Qiong Zhang[†], and Xiaoxiao Li. "Forgettable federated linear learning with certified data removal". *IEEE Transactions on Neural Networks and Learning Systems* (**TNNLS**), 2026.
14. Qiong Zhang*, Yan Shuo Tan*, and Jiahua Chen. "Byzantine-tolerant distributed learning of finite mixture models". **JRSSB**, 2026.
15. Zijian Wang[◇], Xiaofei Zhang, Xin Zhang, Yukun Liu, and Qiong Zhang[†]. "Beyond aggregation: Guiding clients in heterogeneous federated learning". **ICLR**, 2026 (28.18% acceptance).
16. Tao Wang*[,◇], Mengyu Li*[,◇], Geduo Zeng[◇], Cheng Meng[†], and Qiong Zhang[†]. "Gaussian herding across pens: An optimal transport perspective on global Gaussian reduction for 3DGS". **NeurIPS**, 2025 ***(spotlight, 3.19% acceptance)***.
17. Jing Peng[◇], Meiqi Yang, Qiong Zhang[†], Xiaoxiao Li. "S4M: S4 for multivariate time series forecasting with missing values". **ICLR**, 2025 (31.73% acceptance).
18. Qiong Zhang, Archer Gong Zhang, and Jiahua Chen. "Gaussian mixture reduction with composite transportation divergence". *IEEE Transactions on Information Theory* **(TIT)** 70(7), 5191–5212, 2024.
19. Qiong Zhang and Jiahua Chen. "Distributed learning of finite Gaussian mixtures". *Journal of Machine Learning Research* (**JMLR**) 23(1), 4265–4304, 2022.
20. Qiong Zhang and Jiahua Chen. "Minimum Wasserstein distance estimator under finite location-scale mixtures". In *Advances and Innovations in Statistics and Data Science*, pp. 69–98. Springer, Cham, 2022.
21. Qiong Zhang and Jihua Chen. "Robustness of Gaussian mixture reduction for split-and-conquer learning of finite Gaussian mixtures". 3rd *International Conference on Statistics: Theory and Applications* (ICSTA), 2021.
22. Hanwen Liang*, Qiong Zhang*, Peng Dai, and Juwei Lu. "Boosting the generalization capability in cross-domain few-shot learning via noise-enhanced supervised autoencoder". *International Conference on Computer Vision* (**ICCV**), 2021 (25.9% acceptance).
23. Xin Ding*, Qiong Zhang*, and William J. Welch. "Classification beats regression: Counting of cells from greyscale microscopic images based on annotation-free training samples". *CAAI International Conference on Artificial Intelligence*, 2021 (34.5% acceptance).
24. Zhanshou Chen, Jiahua Chen, and Qiong Zhang. "Small area quantile estimation via spline regression and empirical likelihood". *Survey Methodology* 45(1), 81–99, 2019.
25. Philippe Phan, Brandon Budhram, Qiong Zhang, Carly S. Rivers, Vanessa K. Noonan, Tova Plashkes, Eugene K. Wai, Jérôme Paquet, Darren M. Roffey, Eve Tsai, and Nader Fallah. "Highlighting discrepancies in walking prediction accuracy for patients with traumatic spinal cord injury: An evaluation of validated prediction models using a Canadian multicenter spinal cord injury registry". *The Spine Journal*, 19(4), 703–710, 2019.
26. Bo Chang*, Qiong Zhang*, Shenyi Pan, and Lili Meng. "Generating handwritten Chinese characters using CycleGAN". In 2018 IEEE *Winter Conference on Applications of Computer Vision* (**WACV**), pp. 199–207. IEEE, 2018 (45.9% acceptance).

Technical Reports

27. Ruinan Jin, Gexin Huang, Xinwei Shen, Qiong Zhang, Yan Shuo Tan, and Xiaoxiao Li. "Reference image-guided comparative vision-language models for medical diagnosis". Available on arXiv.
28. Qiong Zhang, Jing Peng*[,◇], Xin Zhang*, Aline Talhouk, Gang Niu, and Xiaoxiao Li. "FedMT: Federated learning with mixed-type labels". Available on arXiv.

Grants & Awards

Grants

- 2025–2029 **National Key R&D Program of China – Young Scientists Project (Co-PI)**
- 2024–2026 Key Laboratory of Advanced Theory and Application in Statistics and Data Science Open Research Fund (PI)
- 2024–2026 **National Natural Science Foundation of China Young Scientists Fund (PI)**
- 2023–2025 Renmin University of China (RUC) Startup Research Grant (PI)
- 2023–2024 RUC Early Development Research Grant (PI)

Awards

- 2026 CVPR Outstanding Reviewer Award
- 2026 NeurIPS Top AC Award
- 2026 ICML Silver Reviewer Award
- 2021 Honorable Mention for the Presentation Award, 2nd Waterloo Student Conference in Statistics, Actuarial Science and Finance
- 2019 Winner of SSC (Statistical Society of Canada) Annual Meeting Case Study
- 2017 Margaret Wylie Memorial Scholarship in Statistics
- 2017–2021 University of British Columbia (UBC) International Doctoral Fellowship
- 2017–2021 UBC Faculty of Science Graduate Award
- 2016 CANSSI (Canadian Statistical Sciences Institute) Scholarship
- 2015–2021 UBC International Tuition Award
- 2013, 2014 University of Science and Technology of China (USTC) Outstanding Undergraduate Scholarship
- 2011 USTC Outstanding Freshman Scholarship

Teaching

Instructor, Renmin University of China
Undergraduate Level Courses

▪ Introduction to probability and mathematical statistics	09/2025–12/2025

Master's Level Courses

▪ Data visualization	09/2025–12/2025

Ph.D. Level Courses

▪ Applied statistics	03/2026–06/2026
▪ Bayesian modeling and inference	02/2025–06/2025 02/2024–06/2024 02/2023–06/2023
▪ Special topics in big data (with a focus on distributed learning)	02/2025–06/2025 02/2024–06/2024 02/2023–06/2023

- Advanced statistical computing — 09/2025–12/2025, 09/2024–12/2024, 09/2023–12/2023, 09/2022–12/2022

Teaching Assistant, University of British Columbia
Held weekly labs and office hours, created and marked assignments and exams

- STAT 201: Statistical inference for data science — 01/2022–04/2022
- STAT 404: Design and analysis of experiments — 09/2021–12/2021
- STAT 305: Introduction to statistical inference — 07/2021–08/2021
- STAT 251: Elementary statistics — 05/2021–06/2021
- STAT 300: Intermediate statistics for applications — 01/2021–04/2021
- STAT 344: Sample surveys — 09/2020–12/2020
- STAT 302: Introduction to probability — 01/2020–04/2020, 09/2019–12/2019
- STAT 461/561: Statistical theory II — 01/2019–04/2019
- STAT 306: Finding relationships in data — 09/2018–12/2018
- STAT 200: Elementary statistics for applications — 01/2018–04/2018, 09/2017–12/2017, 01/2017–04/2017, 09/2016–12/2016, 01/2016–04/2016, 09/2015–12/2015

Teaching Assistant, University of Science and Technology of China
Held weekly TA office hours, marked assignments and exams

- Linear algebra (B1) — 02/2015–06/2015
- Linear algebra (B2) — 09/2014–01/2015

Other

- Trainer for teaching assistant program (@ UBC Statistics) — 09/2019–09/2021
- UBC Instructional Skills Workshops — 11/2019

Talks

Invited Talks

- 09/2026 CAS@AMSS: A two-way street: How statistical thinking powers AI efficiency and how AI inspires new statistical inference?
- 07/2026 CSIAM-BDAI 2026: A two-way street: How statistical thinking powers AI efficiency and how AI inspires new statistical inference?
- 06/2026 USTC@CS: Beyond accuracy: Statistical foundations for efficient and reliable AI.
- 06/2026 2026 Annual Academic Conference of the Multivariate Analysis Application Professional Committee of the China Association for Applied Statistics: TabClustPFN: A prior-data fitted network for tabular data clustering.

- 05/2026 Xiamen University@WISE: A two-way street: How statistical thinking powers AI efficiency and how AI inspires new statistical inference?
- 04/2026 NTU@SPMS: A two-way street: How statistical thinking powers AI efficiency and how AI inspires new statistical inference?
- 04/2026 NUS@DSDS: A two-way street: How statistical thinking powers AI efficiency and how AI inspires new statistical inference?
- 01/2026 MBZUAI: A two-way street: How statistical thinking powers AI efficiency and how AI inspires new statistical inference?
- 08/2025 RIKEN@AIP: Robust and scalable federated learning of finite mixture models: Addressing label switching and Byzantine failures.
- 08/2025 Gouxionghui: Distributed learning of finite mixture models with and without Byzantine failures.
- 08/2025 EcoSta 2025: Byzantine tolerant distributed learning of finite mixture models.
- 06/2025 Yunnan University Conference: TabPFN: One model to rule them all?
- 12/2024 NENU Seminar: Distributed learning of finite mixture models with and without Byzantine failures.
- 06/2024 USTC Seminar: Distributed learning of finite mixture models with and without Byzantine failures.
- 06/2024 East China Normal University Seminar: Distributed learning of finite mixture models with and without Byzantine failures.
- 12/2023 Banff International Research Station (BIRS)–Institute for Advanced Study in Mathematics (IASM) Workshop on Harnessing the Power of Latent Structure Models and Modern Big Data: Distributed learning of finite mixture models.
- 12/2023 East China Normal University Colloquium: Gaussian mixture reduction with composite transportation divergence.
- 11/2023 Nankai University Seminar: Distributed learning of finite Gaussian mixtures.
- 08/2023 1st International Conference on Machine Learning and Statistics: Gaussian mixture reduction with composite transportation divergence.
- 05/2023 USTC Seminar: Distributed learning of finite Gaussian mixtures.
- 12/2022 Xiamen University Conference: Gaussian mixture reduction with composite transportation divergence.
- 11/2022 Shanghai Jiao Tong University Seminar: Distributed learning of finite Gaussian mixtures.
- 10/2022 Renmin University of China Conference: Federated learning with mixed-type labels.

Contributed Talks

- 08/2024 JSM: Byzantine tolerant distributed learning of finite mixture models.
- 07/2023 Joint Conference on Statistics and Data Science in China (JCSDS): Gaussian mixture reduction with composite transportation divergence.
- 2021 Waterloo Student Conference of Statistics, Actuarial Science and Finance: Distributed learning of finite Gaussian mixtures.
- 2021 JSM: Distributed learning of finite Gaussian mixtures.
- 2021 ICSTA: Robustness of Gaussian mixture reduction for split-and-conquer learning of finite Gaussian mixtures.
- 2021 UBC/SFU Joint Student Seminar: Distributed learning of finite Gaussian mixtures.
- 2018 UBC/SFU Joint Student Seminar: Generating handwritten Chinese characters using CycleGAN.

- 08/2016 Statistics Canada: Estimation of small area means and quantiles using EBLUP, Pseudo-EBLUP and M-quantile approaches.

Poster Presentations

- 2025 NeurIPS: Gaussian herding across pens: An optimal transport perspective on global Gaussian reduction for 3DGS.
- 2023 IMS New Researcher Conference: Distributed learning of finite Gaussian mixtures.
- 2022 NeurIPS Journal-to-Conference Track: Distributed learning of finite Gaussian mixtures.
- 2021 CANSSI Showcase: Distributed learning of finite Gaussian mixtures.
- 2019 SSC (Statistical Society of Canada) Case Study: Classification beats regression in cell counting from microscopic images.
- 2018 JSM Data Expo: Do I really need a jacket?
- 2018 WACV: Generating handwritten Chinese characters using CycleGAN.

Professional Experience & Activities

Student Supervision

- Tianqi Zhao (Ph.D.) 2025–present
- Yimei Zhang (Ph.D.) 2024–present
- Zijian Wang (Ph.D., co-advised) 2024–2026
- Jing Peng (MSc.) → Georgia Institute of Technology (CS Ph.D.) 2023–2025
- Jianhuang Gan (MSc.) → Fuwai Hospital 2022–2024
- Pengcheng Kong (MSc.) → Shanghai Stock Exchange 2022–2024
- Cong Ye (MSc.) → Iowa State University (Stat Ph.D.) 2022–2024

Service

- NeurIPS Area Chair
- Graduate Committee, Institute of Statistics and Big Data, RUC 2023–2026
- Faculty Search Committee, Department of Statistics, UBC 2022

Reviewer

- Statistics: Statistica Sinica, Journal of Multivariate Analysis, Electronic Journal of Statistics, Journal of the American Statistical Association
- Machine Learning: Machine Learning, IEEE Transactions on Neural Networks and Learning Systems, Journal of Machine Learning Research, International Conference on Machine Learning, International Conference on Learning Representations, Neural Information Processing Systems, Transactions on Machine Learning Research
- Computer Vision: IEEE Transactions on Image Processing, Conference on Computer Vision and Pattern Recognition
- General: Scientific Reports

Organizer & Conference Volunteer

- RUC ISBD Department Seminar Organizer 2023–2024

- UBC Constance van Eeden Lecture Organizer 2019–2020
- UBC/SFU Joint Student Seminar Organizer 2017–2019
- 2018 JSM–ICSA Volunteer 08/2018
- ICSA-Canada Chapter 2017 Symposium Volunteer 08/2017

Internship

- Huawei Noah's Ark Lab, Markham, ON, Canada 05/2020–09/2020
  Computer Vision Team
- Rick Hansen Institute, Vancouver, BC, Canada 05/2017–08/2017
- Statistics Canada, Ottawa, ON, Canada 06/2016–08/2016
  International Cooperation and Corporate Statistical Methods Division